

Elements of Statistics and Probability

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1 Probability

Probability

- Probability is a number between 0 and 1 that reflects the chance or likelihood that a particular event will occur
- Probabilities can be expressed as proportions that range from 0 to 1 or as percentages ranging from 0% to 100%

Terminologies

- Some keywords used while studying probability:
 - Experiment
 - Trial
 - Outcome
 - Sample space
 - Event

Experiment

- An experiment is a process that can be repeated several times under same conditions leading to one of several possible outcomes. We can't tell in advance what the outcome is going to be in each trial even if we can specify the possible outcomes
- In short, An experiment is a process that can be repeated several times under same conditions and have a well defined set of outcomes
- Experiment may be **deterministic** or predictable and **Random** or unpredictable

Deterministic or Predictable Experiment

- An experiment is called **deterministic** when the outcome or result is unique or certain. Everyone conducting the experiment will get the same result or outcome
- Examples:
 - 1 Predicting the amount of money in a bank account if you know the initial deposit and the interest rate
 - 2 The relationship between a circumference and radius of a circle, or the area and radius of a circle

Random or Unpredictable Experiment

- An experiment whose outcomes can not be predicted with certainty in advance is called random or unpredictable experiment
- Examples:
 - 1 An experiment of tossing a coin
 - 2 An experiment of rolling a die
 - 3 Drawing a card from a pack

Trial

- A trial is a physical action, the result of which cannot be predetermined

Outcome

- A possible result of a trial of an experiment

Sample Space

- A set of all possible outcomes of an experiment and it is denoted by S
- Each outcome of an experiment is a **sample point** or element in the sample space

Sample Space (cont.)

- Examples:

- 1 Tossing a coin: $S = \{H, T\}$ (two sample points)
- 2 Throwing a dice: $S = \{1, 2, 3, 4, 5, 6\}$
- 3 Lifetime of a lightbulb: $S = \{x | 0 \leq x < \infty\} = [0, \infty)$

Sample Space (cont.)

- Consider an experiment that consists of rolling two balanced dice, one white and one red are thrown and number of dots on their upper faces are noted, also if b be the outcomes of the white die and r be the outcomes of the red die. If we let denote the outcome in which white dice has value w and red dice has value r , then the sample space of this experiment is:

Sample Space (cont.)

		White Die					
		1	2	3	4	5	6
Red Die	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Event

- Set of one or more possible outcomes of an experiment
- Any subset of sample space is an event
- For example, in throwing a die experiment, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$
- $E = \{2, 4, 6\}$ is an event, which can be described in words as “the number is even”

Example

- 1 If the experiment consists of flipping a coin, then possible outcomes are Head (H) and Tail(T) and the sample space is : $S = \{H, T\}$
- 2 If the experiment consists of rolling a fair die, then possible outcomes are 1, 2, 3, 4, 5 or 6 and the sample space is : $S = \{1, 2, 3, 4, 5, 6\}$

Exercise

- Define the experiment, trial, outcomes and sample space:
 - 1 Flip two coins
 - 2 Roll two dice
 - 3 Measure the lifetime of a machine

Mutually Exclusive Events

- Two events are called mutually exclusive if the occurrence of one event means that none of the other events can occur simultaneously in a single trial. In other words, if one of those events occur, then the other events will not occur

Example

- In tossing a coin experiment, event $E_1 = \{\text{Head}\}$ and event $E_2 = \{\text{Tail}\}$ are mutually exclusive events as both of the events E_1 & E_2 can not occur at the same time. These are mutually exclusive events
- On a day, Event $E_1 = \{\text{Rain}\}$ & event $E_2 = \{\text{Sunny}\}$ may occur simultaneously. These are not mutually exclusive events

Collectively Exhaustive Events

- Events of an experiment are said to be collectively exhaustive events if they include all possible outcomes

Example

- In a coin tossing experiment event $E_1 = \{\text{Head}\}$ and event $E_2 = \{\text{Tail}\}$ are collectively exhaustive, because together they comprise the all the outcomes that are possible in a coin tossing experiment. There are no other possible outcomes of this experiment than these two

Equally Likely Events

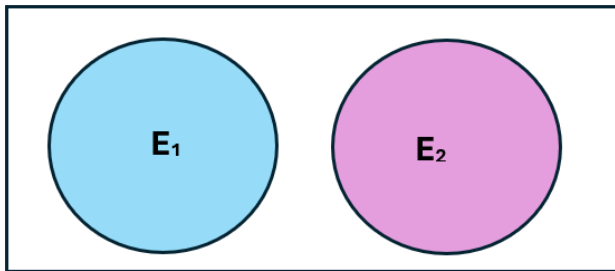
- Equally likely events are events that have the same probability or likelihood of occurring

Example

- Each numeral on a die is equally likely to occur when the die is tossed
- The sample space of throwing a die is, $S = \{1, 2, 3, 4, 5, 6\}$ and the probability of getting a chosen numeral $= \frac{1}{6}$. Here the chance of occurring each numeral is the same and so they are equally likely events

Disjoint Events

- Two events are called disjoint, if they have no common elements between them
- Mutually exclusive events are disjoint events

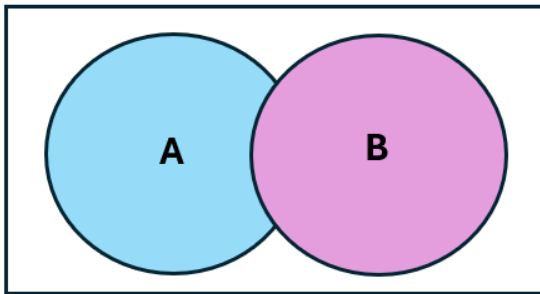


Dependent Events

- If two events have some common elements then both of these events are referred as dependent events
- For example, in a deck of 52 cards, if E_1 be an event of selecting 'Red' card and E_2 be another event of selecting 'Queen' card then these events are joint events since there are two 'Queen' cards in a set of 'Red' cards

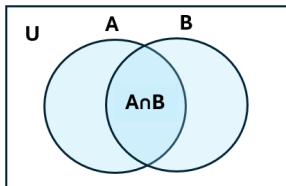
Union

- The union of two sets contains all the elements contained in either set (or both sets). The union is notated $A \cup B$, where A and B are two sets
- $A \cup B$ means: Elements that belong to either A , B or both



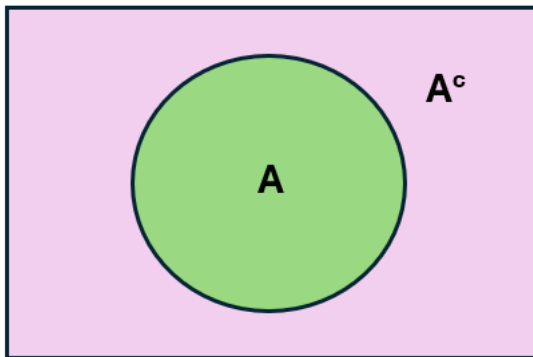
Intersection

- The intersection of two sets contains only the elements that are in both sets. The intersection is notated $A \cap B$
- $A \cap B$ means: Elements that belong to both A & B



Compliment

- The complement of a set A contains everything that is not in the set A . The complement is notated A' , or A^c
- A' means: Elements that don't belong to A



Independent Event

- Two events are known as independent events if the occurrence of one event does not affect the probability of occurring another event
- For two independent events A and B, the probability that A and B will both occur is found by multiplying the two probabilities

$$P(A \cap B) = P(A)P(B)$$

- For example; In a coin tossing experiment events, $E_1 = \{\text{Head}\}$ and event $E_2 = \{\text{Tail}\}$ are independent events because in this case occurrence of event, E_1 does not affect event, E_2

Approaches of Assigning Probability

- Classical approach
- Frequency approach

Classical Approach

- If a random experiment has a total of $n(S)$ possible outcomes, all of which are mutually exclusive, equally likely and collectively exhaustive, such that $n(A)$ of the outcomes are favorable to an event A , then the probability of the event A is defined by

$$P(A) = \frac{n(A)}{n(S)}$$

Frequency Approach

- If an experiment is repeated n times under the same conditions and event A occurs f times out of n times, then

$$P(A) = \lim_{n \rightarrow \infty} \frac{f}{n}$$

- That is, when n is very large, $P(A)$ is very close to the relative frequency of event A

Example

- In a dice throwing experiment, $S = \{1, 2, 3, 4, 5, 6\}$ and our favorable event is $A = \{2\}$
- Let, 2 occurred a total of 998 times out of total 6000 trials.

Therefore,

$$P(A) = \lim_{n \rightarrow \infty} \frac{998}{6000} \approx \frac{1}{6}$$

Three Axioms of Probability

- ① **Axiom 1:** (Axiom of positivities): $0 \leq P(A) \leq 1$
- ② **Axiom 2:** (Axiom of certainty): $P(S) = 1$
- ③ **Axiom 3:** (Axiom of additivity): For a sequence of disjoint events $A_1, A_2, \dots, A_n$:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

Example

- A person holds ticket in a lottery that offers 10 prizes and sells 120 tickets. What is the probability that the person will not win a prize?

- **Solution:**

- Let A be the event of winning a prize
 - Here, $P(A) = \frac{n(A)}{n(S)} = \frac{10}{120} = \frac{1}{12}$
 - Thus, $P(A') = 1 - \frac{1}{12} = \frac{11}{12}$
- So, the probability that the person will not win a prize is $\frac{11}{12}$

Example

- A dice is thrown in an experiment. What is the probability that an even no. will occur?
- **Solution:**
 - The sample space for the experiment is, $S = \{1, 2, 3, 4, 5, 6\}$
 - Let the event is $E = \{2, 4, 6\}$
 - Here, $n(E) = 3$ and $n(S) = 6$
 - Therefore, the probability of occurring the event, E

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

Law of Addition

- For two **disjoint** events A and B:

$$P(A \cup B) = P(A) + P(B)$$

- For **disjoint** events A, B, ..., Z:

$$P(A \cup B \cup \dots \cup Z) = P(A) + P(B) + \dots + P(Z)$$

- For two **joint** events A and B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- For three **joint** events A, B and C:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

Example

- In a company, 60% of the employees have motorcycle, 40% has private car and 20% has both. If an employee is selected randomly from that company, then
 - (a) What is the probability that the employee has either motorcycle or private car?
 - (b) What is the probability that the employee has neither motorcycle nor private car?

Solution

- Let, M = the randomly selected employ has motorcycle
- C = the randomly selected employee has car
- Here, $P(M) = \frac{60}{100}0.6$, $P(C) = \frac{40}{100} = 0.4$ and
$$P(M \cap C) = \frac{20}{100} = 0.2$$

Solution

- (a) Probability that the person has either motorcycle or private car is,

$$\begin{aligned}P(M \cup C) &= P(M) + P(C) - P(M \cap C) \\&= 0.6 + 0.4 - 0.2 \\&= 0.8\end{aligned}$$

- (b) Probability that the person has neither motorcycle nor private car is

$$\begin{aligned}P((M \cup C)^c) &= 1 - P(M \cup C) \\&= 1 - 0.8 \\&= 0.2\end{aligned}$$

Thank You